

Review Aplikasi personal-v4

Laravel 12 + Filament v5 — dashboard personal all-in-one.

Author: Nova Ardiansyah

Ringkasan

Aplikasi dashboard personal yang menggabungkan manajemen keuangan, utang, email, blog, file, monitoring uptime, gallery, notes, short URL, sistem notifikasi (Telegram + Push), dan tracking aktivitas — semua via admin panel Filament SPA + REST API.

1. Fitur yang Ada (Existing Features)

1.1 Finance & Payment System (`app/Models/Payment*` , `app/Filament/Resources/Payments/`)

Status:  Paling matang

- Income / Expense / Transfer / Withdrawal dengan mutasi saldo otomatis ke Payment Account
- Payment Accounts (many-to-many transfer)
- Payment Categories (dengan is_default)
- Draft payments (approval flow, saldo tidak terpotong sampai approve)
- Scheduled payments (auto-proses via queue saat tanggal tiba)
- Itemized transactions (Item + Pivot dengan quantity/price/total)
- Overview report: daily, weekly, monthly expense/income
- Report: saldo total, pengeluaran terjadwal, draft
- PDF report generation via mPDF
- Excel export (via maatwebsite)
- Dashboard widget: chart pengeluaran per kategori + stats overview
- Soft deletes di semua model

1.2 Debt Management (`app/Models/Debt*`)

Status:  Fitur baru, solid

- Debt tracking dengan platform, principal, admin fee, disbursement
- Interest rate + service fee rate
- Annuity payment calculation
- Auto-generate installments (tenor-based)
- Installment status: unpaid / paid
- Payment status: ongoing / partial_payment / paid
- Setiap installment terbayar auto-create Payment record (draft)
- Relasi ke Payment Account

1.3 Blog System (`app/Models/Blog*`)

Status:  **Cukup matang**

- Blog posts dengan categories + tags
- Status: Draft / Published / Scheduled
- Cover image upload
- SEO fields: meta title, meta description
- Blog subscribers (auto email notification)
- Author management (polymorphic user)
- Scheduled publishing

1.4 Email System (`app/Models/Email*`)

Status:  **Mature**

- Email sending dengan Mail queue
- Email Templates dengan placeholder system
- Header/footer toggle
- Attachments via polymorphic File model
- Status tracking: Draft → Sent
- Preview di admin panel
- Auto-login notification email (dengan template berbeda untuk API vs Web)
- UID tracking per email

1.5 Uptime Monitoring (`app/Models/UptimeMonitor*`)

Status:  **Mature**

- HTTP endpoint monitoring
- Response time tracking
- Status: UP / DOWN / SLOW (threshold 300ms)
- Auto-check via queue dengan interval configurable

- Telegram notification on DOWN → UP / UP → DOWN
- HttpStatusCode reference table
- Log per check (total_checks, healthy_checks, unhealthy_checks)

1.6 Gallery ([app/Models/Gallery](#))

Status:  **Mature**

- Image upload with polymorphic subject (attach ke Payment, dll)
- Auto-optimize: small (300px), medium (900px), large (1600px), original
- Image size classification enum
- Group code untuk grouping
- Private/public toggle
- Soft deletes

1.7 Notes ([app/Models/Note](#))

Status:  **Sederhana & fungsional**

- CRUD notes
- Pin / Archive toggle
- File attachments (polymorphic)
- Soft deletes

1.8 Short URLs ([app/Models/ShortUrl](#))

Status:  **Fitur lengkap**

- URL shortener dengan custom domain support
- QR code generation (HeroQR)
- Click tracking
- Note per short URL
- Integrasi dengan File Downloads

1.9 File Management ([app/Models/File](#), [app/Models/FileDownload](#))

Status:  **Mature**

- Polymorphic file attachments (Email, Note, dll)
- Temporary signed download URL (1 month expiry)

- File aliases
- Size tracking
- Scheduled auto-deletion
- Download tracking

1.10 Contact Messages ([app/Models/ContactMessage](#))

Status:  **Sederhana**

- Public API endpoint untuk contact form
- Read/unread status
- Soft deletes
- Reply triggers via Telegram/Email

1.11 Push Notifications ([app/Models/PushNotification](#))

Status:  **Fungsional**

- Expo push notification integration
- Per-user notification token
- Allow/disallow notification toggle
- Sent/failed tracking

1.12 Telegram Notifications ([app/Services/TelegramService](#))

Status:  **Mature**

- Full Telegram bot integration
- Notifications untuk: login, uptime (down/up), debt, payment, activity
- Event-based + Job-based dispatching

1.13 Activity Logging ([app/Models/ActivityLog](#))

Status:  **Mature**

- Resource-level activity logging (Created/Updated/Deleted)
- Device info tracking: IP, country, city, region, geolocation, timezone, user agent, referer
- Polymorphic causer/subject
- Preview email dari activity log

1.14 Authentication & User ([app/Models/User](#) , [app/Services/AuthService](#))

Status:  **Mature**

- Filament multi-factor auth: App Authenticator + Email OTP
- Sanctum API tokens
- Avatar upload + URL
- Login notification (Telegram + Email) dengan dedup interval
- IP geo-location on login (ipinfo.io)
- Soft deletes

1.15 Code Generation ([app/Models/Generate](#))

Status:  **Fitur utility kuat**

- Auto-generate kode sequential per model/alias
- Format: prefix + YYYYMM + queue(0001-9999) + suffix
- Auto-reset per bulan
- Support banyak model: payment, short_url, debt, dll

1.16 Settings ([app/Models/Setting](#))

Status:  **Utility**

- Key-value dengan cache forever
- Setting untuk: short_url_domain, telegram chat_id, interval notif, default_system_user, dll

1.17 Skills ([app/Models/Skill](#))

Status:  **Sederhana**

- Daftar skill (untuk portfolio/showcase)
-

2. API Routes ([routes/api/](#))

Status:  **Terstruktur**

- Auth: login, logout, validate-token, change-password, update-profile, login-notification
- Payments CRUD + report + PDF

- Payment Accounts CRUD
- Payment Types CRUD
- Payment Goals CRUD
- Payment Goal Statuses
- Items & Item Types
- Notes CRUD
- Blog: posts, categories, tags, subscribers
- Gallery CRUD
- Short URLs
- Contact Messages
- Notifications: settings, test, telegram send
- Push Notifications
- Code generation
- Health check endpoint

3. Area yang Perlu Improve

3.1 Konsistensi Pola Kode

Issue	Detail	Saran
Campur static & instance method	<code>Payment::mutateDataPayment()</code> static tp dipanggil instance	Pilih salah satu, refactor ke Service class murni
Logika mutasi saldo duplikasi	<code>Payment::mutateDataPayment()</code> , <code>Payment::approveDraft()</code> , <code>PaymentService::manageDraft()</code> hampir identik	Extract ke service method tunggal
Inline HTML di service	<code>UptimeMonitorService::sendDownNotification()</code> hardcode string	Pindahkan ke view template

3.2 Security Gaps

Issue	Detail	Tingkat
No rate limiting API	Tidak ada throttle di route API publik	Medium
No CSRF di webhook	<code>/webhook/testing</code> tanpa middleware CSRF	Low (POST-only)
Signed URL expiration hardcoded	1 month di <code>makePdf()</code>	Low, tp jadi perhatikan

Issue	Detail	Tingkat
No ownership check detail	Beberapa API query tanpa <code>where('user_id', ...)</code>	Perlu audit per controller

3.3 Performance

Issue	Detail	Saran
Query N+1 potensial	<code>Payment::overviewReport()</code> panggil <code>PaymentAccount::sum()</code> terpisah	Join atau cache saldo
Cache setting unbounded	<code>cache()->rememberForever()</code> untuk setting – tidak pernah invalidate	Tambahkan event listener clear cache saat setting diupdate
Chunk size hardcoded	<code>chunk(200)</code> di PDF, <code>chunkById(50)</code> di uptime	Jadikan configurable

3.4 Kualitas UX & Admin

Issue	Saran
Table/Pencarian	Beberapa resource belum ada global search/filter yang optimal
Bulk actions	Belum ada bulk delete/export di beberapa resource
Validation error handling	<code>normalizeValidationErrors()</code> helper ada tp belum dipakai konsisten

3.5 Testing Coverage (dari struktur folder)

Terlihat `tests/` ada, tp belum dicek isinya. Berdasarkan feature complexity:

- Payment system → butuh comprehensive feature tests
- Debt calculations → butuh unit tests untuk annuity formula
- Uptime checker → butuh test dengan HTTP mock
- API endpoints → butuh integration tests

4. Ide Fitur Baru

4.1 High Priority (High Value, Low Effort)

Fitur	Deskripsi	Value
-------	-----------	-------

Fitur	Deskripsi	Value
Recurring / Subscription Tracker	Track langganan bulanan (Netflix, Spotify, dll) dengan auto-reminder sebelum tagihan	Tinggi – enhancement payment system existing
Budget Planning	Budget per kategori per bulan + alert ketika mendekati limit	Tinggi – langsung pakai kategori yg ada
Personal Dashboard 2.0	Layout grid custom: ringkasan saldo, upcoming bills, recent notes, utang jatuh tempo	Tinggi
Dark mode admin	Filament sudah support, tinggal enable	Rendah effort

4.2 Medium Priority

Fitur	Deskripsi
API Key Management	Generate multi API key dengan scope permission
Webhook Outgoing	Forward event (payment created, debt due) ke webhook URL eksternal
Habit Tracker	Daily habit tracking dengan streak (memanfaatkan Note + ActivityLog)
Investment Portfolio	Track saham/reksadana/kripto with cost basis + current value
Calendar/Planner	Integrasi jadwal, task, scheduled payments dalam satu timeline view
Expense Scan via OCR	Foto struk → auto-extract amount + store ke payment

4.3 Low Priority / Idea

Fitur	Deskripsi
Two-way Telegram Bot	Chatbot untuk cek saldo, catat pengeluaran via Telegram
Family Sharing	Multi-user dengan shared payment accounts
Personal CRM	Track interaksi dengan kontak (mirip contact message tp 2-way)
AI-powered categorization	Auto-categorize payment berdasarkan nama transaksi
Public API docs playground	Scramble + Swagger UI interaktif untuk testing API
Time Tracking / Logbook	Track waktu kerja per project

5. Observasi Arsitektur

Strengths

- **Polymorphic relations** dipakai efektif (Files, Galleries, ActivityLog) → skalabel
- **Code generation system** rapi dengan sequential per model + auto-reset
- **Service layer** dipisah per resource (walau belum konsisten)
- **Observer pattern** via `# [ObservedBy]` — modern PHP 8 attribute
- **API routes split** per file → maintainable

Weaknesses

- **Duplikasi logika mutasi saldo** di 3 tempat (model static, approveDraft, PaymentService)
 - **Inline HTML + hardcoded strings** di service layer
 - **Belum ada FormRequest** — validasi campur di controller
 - **Cache invalidation** untuk Setting tidak ada mekanisme auto-clear
 - **Test gap** — fitur kompleks (debt, payment, uptime) belum terjamin
-

6. Kesimpulan

Overall: Aplikasi solid dengan fitur personal finance sebagai core strength.

Sudah punya ekosistem yang matang: notifikasi multi-channel, file management, activity logging, dan queue processing. Struktur kode rapi dengan pattern modern (Observers, Service classes, polymorphic relations).

Prioritas utama:

1. Refactor duplikasi logika mutasi saldo
2. Cache invalidation untuk Settings
3. Rate limiting API + audit ownership check
4. Testing coverage untuk fitur finansial

Quick win fitur:

- Recurring subscription tracker (enhance payment is_scheduled)
 - Budget planning per kategori
 - Dashboard ringkasan yang lebih informatif
-

Review generated 2026-07-13 — berdasarkan eksplorasi kode structure.